

IFCS 2026 Data Challenge

Profiling and Predicting Financial Distress in Italian SMEs



Context

You are given the FY2023 financial statements of roughly 14,000 Italian small and medium enterprises (SMEs). Your mission is twofold:

1. **Profile** these firms: what kinds of companies are out there, and how do they differ financially?
2. **Flag** the ones in financial distress before it is too late.

Both tasks matter to lenders, policymakers, and the firms themselves.

The data

One row per SME, 14 financial and structural variables plus the binary target **Financial distress** (TRUE / FALSE). All monetary values are in **thousand euros (k€)**. Geography is available at the Italian **province** level. Full variable definitions are in the accompanying **data dictionary**.

Task A: Clustering (unsupervised)

The objective of this task is to identify groups of firms with similar financial characteristics using clustering techniques. You should explain and justify the selection of the financial variables included in the analysis, as well as any preprocessing or scaling methods applied to the data. The analysis should also include a rationale for the chosen number of clusters and provide an economic interpretation of the characteristics that distinguish each cluster. Finally, examine how the identified clusters are distributed across Italian regions and discuss any notable geographic patterns or differences.

Task B: Classification (supervised)

The objective of this task is to develop a predictive model for **Financial distress** using the available financial variables as predictors. During the challenge, participants will be provided with a `test_features.csv` file containing the financial information for the test firms.

Deliverables

During the challenge, the team leader will receive an email containing a link to a dedicated submission folder. Before the submission deadline, each team must upload the following two files:

1. A slide deck for the **5-minute presentation** of your findings. The presentation should describe the chosen clustering procedure, the resulting clusters and their interpretation, and the geographical relationships revealed by the partition.
2. A file named **predictions.csv** containing your predictions for the **Financial distress** variable for all firms in the test dataset (**test_features.csv**). The file must contain exactly two columns, **Company ID** and **pred_class**, with one row for each firm in the test dataset. The values in **pred_class** must be either **TRUE** or **FALSE**.

Good luck and remember: interpretation counts as much as prediction!